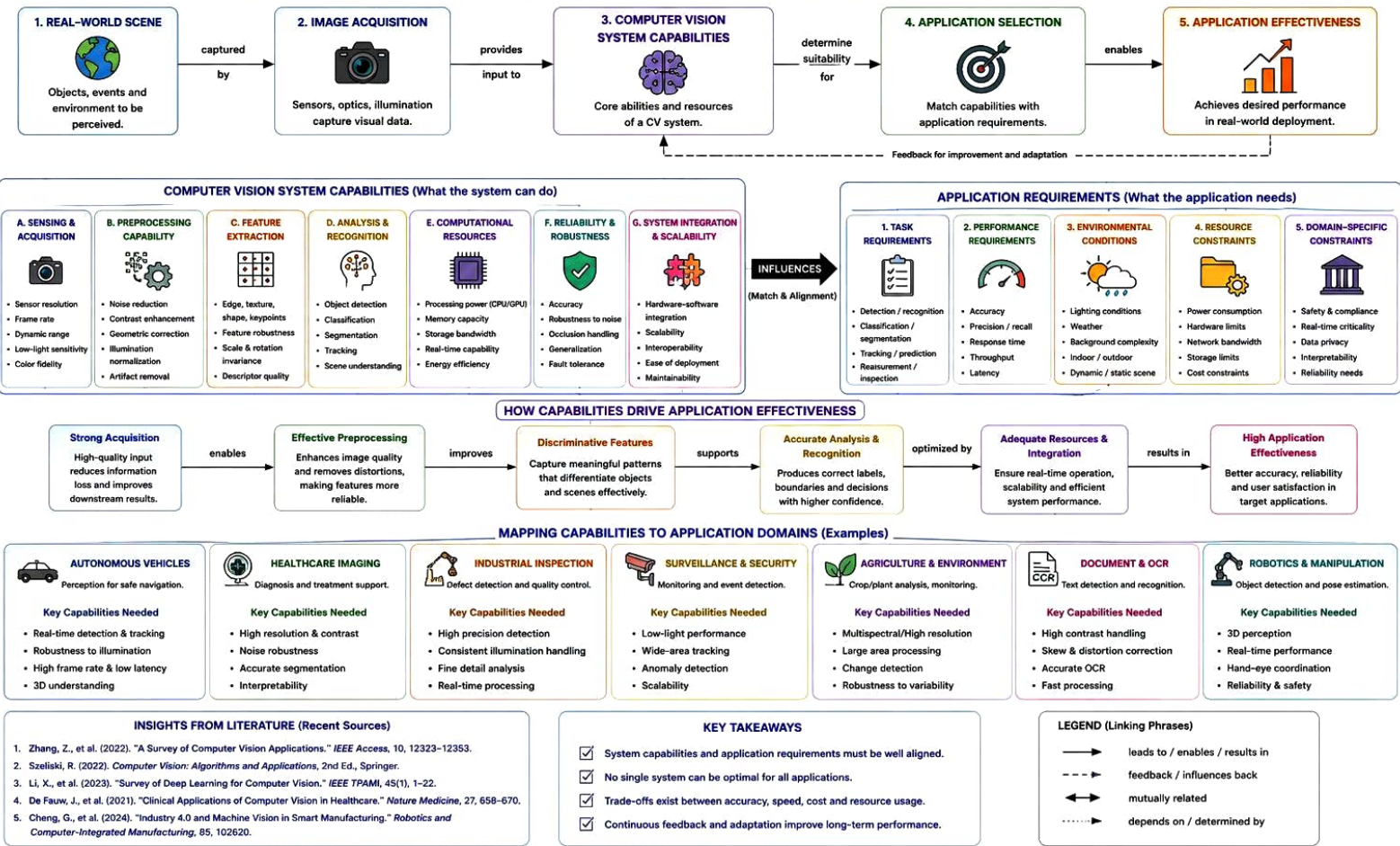


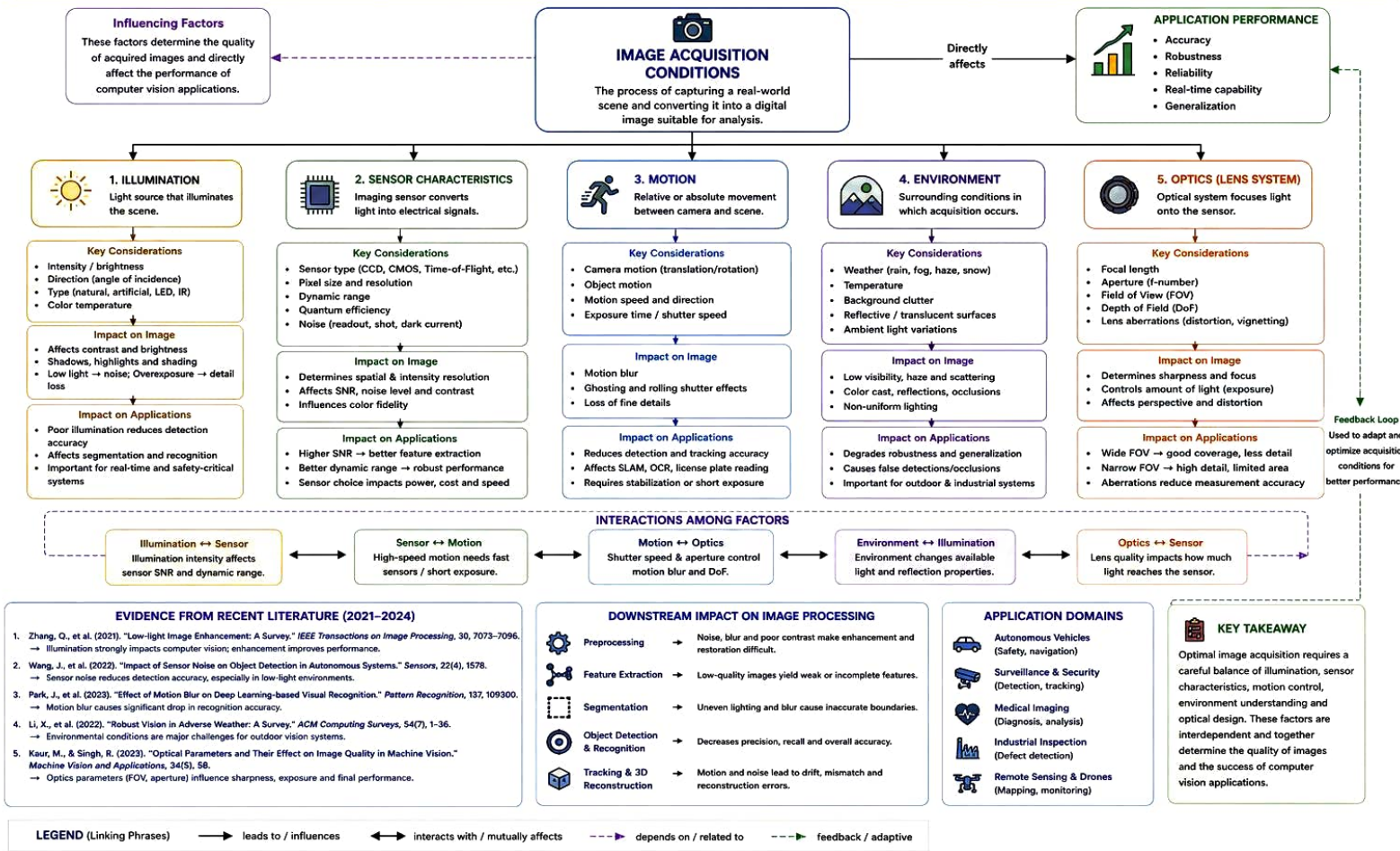
RELATIONSHIP BETWEEN COMPUTER VISION SYSTEM CAPABILITIES AND THE SELECTION OF APPROPRIATE VISION-BASED APPLICATIONS: A CONCEPT MAP

How system capabilities influence application effectiveness – integrating perspectives from multiple literature sources



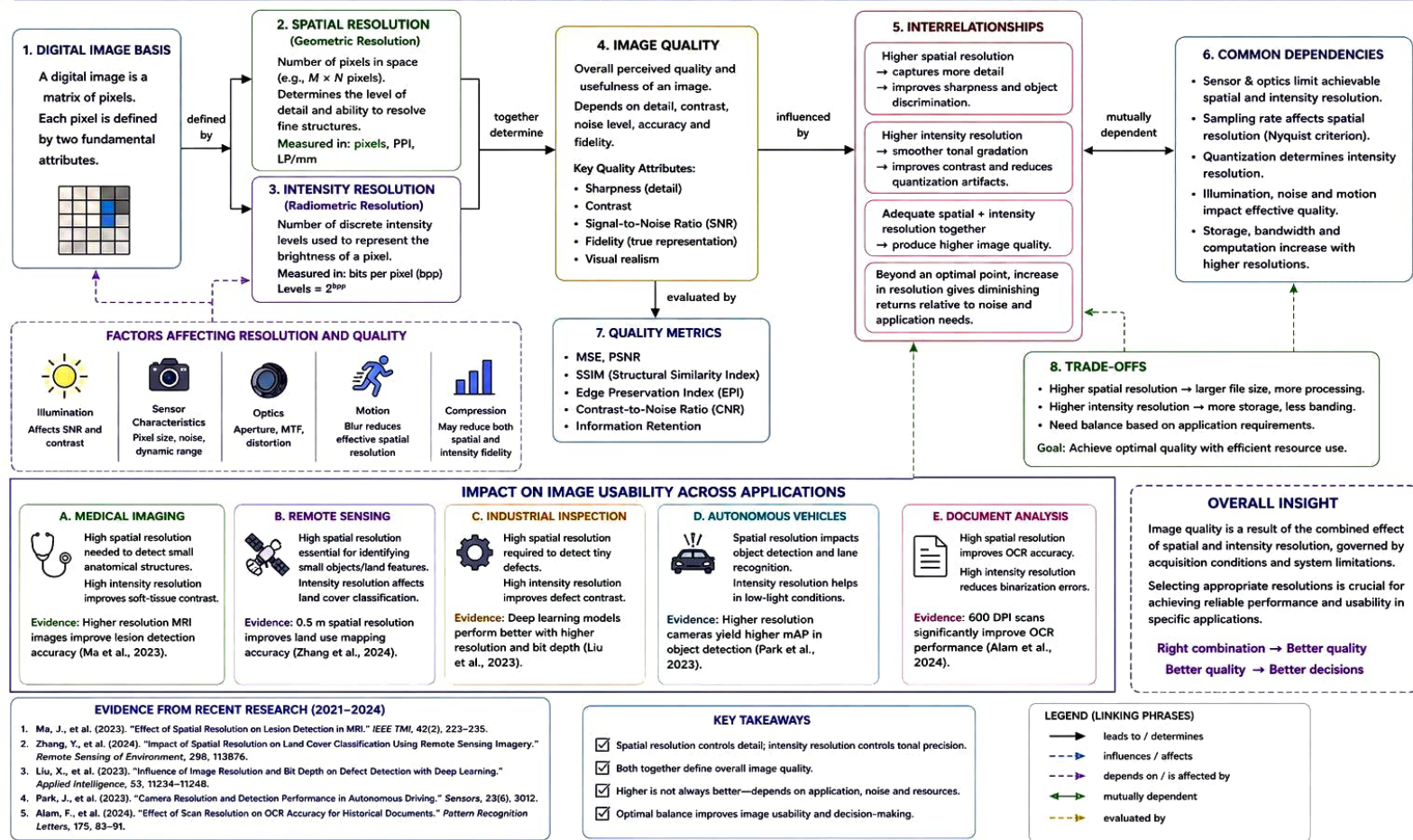
FACTORS INFLUENCING IMAGE ACQUISITION CONDITIONS AND THEIR IMPACT ON APPLICATION PERFORMANCE: A CONCEPT MAP

Showing how illumination, sensor characteristics, motion, environment and optics affect downstream image processing and application performance



RELATIONSHIPS AMONG IMAGE RESOLUTION, SPATIAL RESOLUTION, INTENSITY RESOLUTION AND IMAGE QUALITY: A CONCEPT MAP

Cause-Effect Relationships, Impact on Image Usability in Different Applications and Evidence from Recent Research



FACTORS AFFECTING RESOLUTION AND QUALITY



Illumination
Affects SNR and contrast



Sensor Characteristics
Pixel size, noise, dynamic range



Optics
Aperture, MTF, distortion



Motion
Blur reduces effective spatial resolution



Compression
May reduce both spatial and intensity fidelity

IMPACT ON IMAGE USABILITY ACROSS APPLICATIONS

A. MEDICAL IMAGING



High spatial resolution needed to detect small anatomical structures.
High intensity resolution improves soft-tissue contrast.

Evidence: Higher resolution MRI images improve lesion detection accuracy (Ma et al., 2023).

B. REMOTE SENSING



High spatial resolution essential for identifying small objects/land features.
Intensity resolution affects land cover classification.

Evidence: 0.5 m spatial resolution improves land use mapping accuracy (Zhang et al., 2024).

C. INDUSTRIAL INSPECTION



High spatial resolution required to detect tiny defects.
High intensity resolution improves defect contrast.

Evidence: Deep learning models perform better with higher resolution and bit depth (Liu et al., 2023).

D. AUTONOMOUS VEHICLES



Spatial resolution impacts object detection and lane recognition.
Intensity resolution helps in low-light conditions.

Evidence: Higher resolution cameras yield higher mAP in object detection (Park et al., 2023).

E. DOCUMENT ANALYSIS



High spatial resolution improves OCR accuracy.
High intensity resolution reduces binarization errors.

Evidence: 600 DPI scans significantly improve OCR performance (Alam et al., 2024).

OVERALL INSIGHT

Image quality is a result of the combined effect of spatial and intensity resolution, governed by acquisition conditions and system limitations.
Selecting appropriate resolutions is crucial for achieving reliable performance and usability in specific applications.

Right combination → Better quality
Better quality → Better decisions

EVIDENCE FROM RECENT RESEARCH (2021–2024)

- Ma, J., et al. (2023). "Effect of Spatial Resolution on Lesion Detection in MRI." *IEEE TMI*, 42(2), 223–235.
- Zhang, Y., et al. (2024). "Impact of Spatial Resolution on Land Cover Classification Using Remote Sensing Imagery." *Remote Sensing of Environment*, 298, 113876.
- Liu, X., et al. (2023). "Influence of Image Resolution and Bit Depth on Defect Detection with Deep Learning." *Applied Intelligence*, 53, 11234–11248.
- Park, J., et al. (2023). "Camera Resolution and Detection Performance in Autonomous Driving." *Sensors*, 23(6), 3012.
- Alam, F., et al. (2024). "Effect of Scan Resolution on OCR Accuracy for Historical Documents." *Pattern Recognition Letters*, 175, 83–91.

KEY TAKEAWAYS

- ✓ Spatial resolution controls detail; intensity resolution controls tonal precision.
- ✓ Both together define overall image quality.
- ✓ Higher is not always better—depends on application, noise and resources.
- ✓ Optimal balance improves image usability and decision-making.

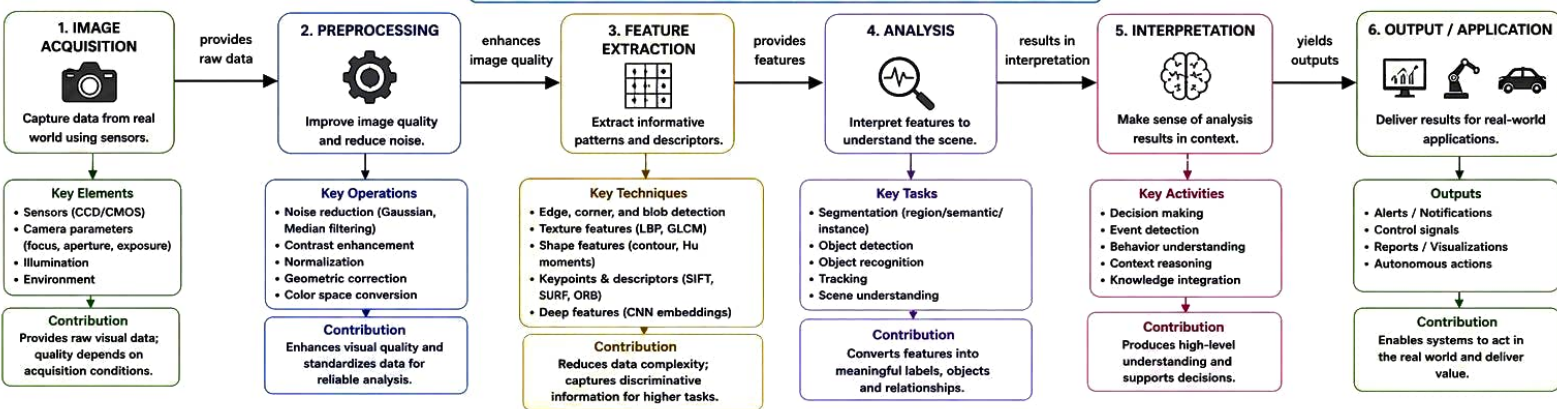
LEGEND (LINKING PHRASES)

- leads to / determines
- influences / affects
- - - depends on / is affected by
- ↔ mutually dependent
- - - evaluated by

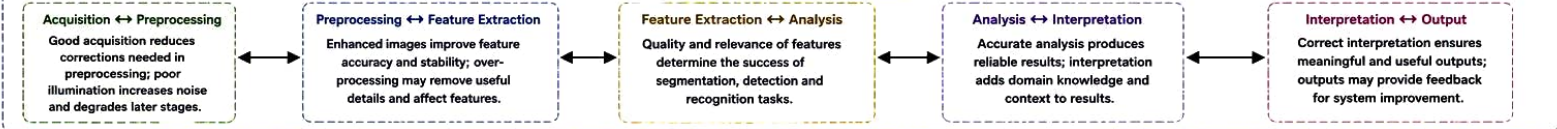
VISUAL DATA HANDLING IN A COMPUTER VISION SYSTEM: STAGES, RELATIONSHIPS AND CONTRIBUTIONS

Objective: To illustrate the major stages of visual data handling, their relationships, contributions to meaningful outputs, and insights from multiple literature sources.

COMPUTER VISION SYSTEM – VISUAL DATA HANDLING PIPELINE



RELATIONSHIPS AMONG STAGES



CONTRIBUTION OF STAGES TO MEANINGFUL OUTPUTS



OVERALL IMPACT

A well-designed pipeline ensures accuracy, robustness, efficiency and scalability, leading to intelligent systems capable of solving complex real-world problems.

INSIGHTS FROM LITERATURE (Recent Key Sources)

- Szeliski, R. *Computer Vision: Algorithms and Applications*, 2nd Ed., Springer, 2022. (Comprehensive explanation of stages from acquisition to interpretation.)
- Gonzalez, R. C. & Woods, R. E. *Digital Image Processing*, 4th Ed., Pearson, 2018. (Details preprocessing and feature extraction foundations.)
- Forsyth, D. A. & Ponce, J. *Computer Vision: A Modern Approach*, 2nd Ed., Pearson, 2021. (Covers the complete pipeline and relationships among modules.)
- Sonka, M., Hlavac, V., & Boyle, R. *Image Processing, Analysis, and Machine Vision*, 5th Ed., Cengage, 2022. (Discusses system design and contribution of each stage.)
- Jain, A. K., Mao, J., & Mohiuddin, K. M. *Artificial Intelligence in Computer Vision*, CRC Press, 2023. (Highlights application-driven perspective and system outputs.)

QUALITY FACTORS AFFECTING ALL STAGES

- Illumination
 - Sensor quality and resolution
 - Noise and distortions
 - Occlusion and clutter
 - Object variability
 - Computational resources
- These factors influence performance at every stage and the final outputs.

FEEDBACK & ITERATION (DEPENDENCIES)

- Results/outputs may be used to:
- Adjust acquisition (camera settings, lighting)
 - Tune preprocessing parameters
 - Select better features
 - Improve models for analysis
- This closed-loop improves overall system performance.

EXAMPLE APPLICATION DOMAINS

- Healthcare (medical image analysis)
- Autonomous Vehicles
- Surveillance & Security
- Industrial Inspection
- Remote Sensing & GIS
- Robotics

LEGEND (Linking Phrases) → leads to / results in --> depends on / influences ↔ interacts with / has mutual influence - - - feedback / iterative

FROM REAL-WORLD SCENE CAPTURE TO DIGITAL IMAGE REPRESENTATION: A CONCEPT MAP

A complete pipeline showing major concepts, relationships, intermediate transformations, dependencies and literature insights

